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Comparative Effectiveness of Exercise Interventions for Hamstring Injury Prevention in Football Players: A Systematic Review and Network Meta-analysis

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Abstract

Background Hamstring strain injury is a leading time-loss injury in football, yet trials have evaluated diverse exercise-based prevention programs with limited direct head-to-head evidence. This systematic review and network meta-analysis aimed to compare the effectiveness of exercise-based interventions for preventing hamstring injuries in football players using exposure-adjusted incidence rate ratios.

Methods We conducted a PRISMA-aligned systematic review and random-effects network meta-analysis of randomized controlled trials in football players comparing exercise-based interventions to prevent hamstring injuries. Searches covered PubMed, Cochrane Library, Embase and Web of Science from inception to January 5, 2026. Interventions were modeled as separate nodes (Nordic hamstring exercise [NHE], FIFA 11, FIFA 11 + , FIFA 11 + Kids, bounding exercise program [BEP], usual training). The primary effect measure was incidence rate ratio (IRR) versus usual training, with ranking by *P*-scores; exploratory subgroup analyses were conducted by sex-restricted male-only evidence and age group.

Results Eleven randomized controlled trials involving 9,282 participants were included. The network was connected but largely star-shaped, with most evidence comparing active interventions against usual training and no direct active–active comparisons. FIFA 11 + showed the most consistent exposure-adjusted association with reduced hamstring injury incidence versus usual training (IRR = 0.55, 95% CI 0.31–0.98; *P*-score = 0.84). FIFA 11 + Kids showed a favorable but imprecise estimate, whereas NHE showed a modest non-significant reduction and FIFA 11 and BEP showed no clear benefit. Male-only findings were broadly consistent with the primary analysis. Age-stratified analyses suggested that the apparent hierarchy was not fully identical across age strata, but subgroup networks were sparse and exploratory. RoB 2 judged three studies as high risk and eight as having some concerns; no trial was judged as overall low risk. Between-study heterogeneity was substantial, and treatment rankings were, therefore, interpreted cautiously.

Conclusions In this IRR-based network meta-analysis, FIFA 11 + showed the most consistent exposure-adjusted association with reduced hamstring injury incidence compared with usual training. However, certainty across the primary comparisons ranged from low to very low because of clinical and methodological heterogeneity, sparse direct evidence for several nodes, reliance on indirect active–active comparisons, substantial between-study heterogeneity, and moderate-to-high risk of bias across the included studies. The treatment hierarchy should, therefore, be

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interpreted as a low-to-very-low-certainty, population-level summary rather than as a basis for firm recommendations. Further well-reported, age- and population-specific head-to-head trials with consistent exposure, injury-definition, adherence, and intervention-dose reporting are needed.

Registration Systematic review registration PROSPERO (CRD420251243576).

Keywords Hamstring injury prevention, Football (soccer), FIFA 11 +, Nordic hamstring exercise, Network meta-analysis

Introduction

Football (i.e., soccer) is a global sport, boasting hundreds of millions of registered players. Consequently, even a modest alteration in the risk of injury can significantly affect both the population level and performance [1, 2]. Hamstring injury represents one of the most prevalent and burdensome non-contact muscle injuries in football, having long been a significant contributor to time-loss injuries across various competitive levels and age groups [3]. Importantly, long-term surveillance data from elite football populations indicate that, despite increasing awareness and implementation of injury-prevention strategies, the incidence and overall burden of hamstring injuries have not shown a consistent downward trend in recent years, particularly in the context of intensified match demands and congested competition schedules [4, 5]. From a team perspective, this translates into multiple hamstring strain injuries (HSIs) per squad each season and a meaningful number of days lost from training and match availability [3, 6]. Greater injury burden has been linked with poorer team performance across a season in professional football [7]. With match play becoming more intense and schedules more congested, preventing HSIs remains a high-priority problem for clinicians, coaches, and performance staff [6, 8, 9].

Most HSIs in football occur during high-speed actions such as sprinting, where the hamstrings experience high mechanical strain and eccentric demands, so prevention efforts often focus on increasing tissue "capacity" to tolerate these loads [10–12]. Prospective cohort data in elite football indicate that short biceps femoris long-head fascicles and low eccentric knee-flexor strength are associated with several-fold higher risk of future HSI [13, 14]. These findings matter clinically, because both architecture and strength respond to training, and different exercises may bias adaptations toward knee-dominant (Nordic-type) versus hip-dominant posterior-chain patterns [14–16]. These individual-level prognostic factors also have implications for comparative effectiveness, because hamstring injury-prevention programs are unlikely to operate through identical mechanisms. Nordic hamstring exercise (NHE) is closely aligned with eccentric knee-flexor strength and hamstring architectural adaptations, whereas FIFA 11 + provides a broader neuromuscular warm-up stimulus targeting strength, balance, trunk–hip control, landing, cutting, and running-related movement quality.

Among specific exercise options, eccentric hamstring strengthening using the NHE has the strongest randomized evidence, with cluster RCTs reporting fewer hamstring injuries when NHE is added to usual football training [17, 18]. Systematic reviews and meta-analyses broadly support these trial findings, reporting that NHE-inclusive programs can reduce hamstring injury rates by around half (or more), particularly when compliance is adequate [5, 19]. Despite this, surveys and implementation studies suggest that NHE is still not routinely adopted in elite football, even when medical and coaching staff are aware of the evidence [4, 19, 20]. Feasibility barriers—such as soreness, perceived fatigue cost, and competing training priorities—have motivated research into dosing strategies and modified variants (e.g., reduced-volume or assisted NHE) designed to preserve adaptations while improving tolerability [21, 22]. In parallel, multi-component neuromuscular warm-up programs such as FIFA 11 + have demonstrated reductions in overall injury incidence in cluster RCTs across different football populations and are designed to improve lower-limb strength, balance, and movement control [2, 23–26]. In collegiate men's soccer specifically, a secondary analysis of the FIFA 11 + trial suggests that the program can also reduce hamstring strain incidence, with better outcomes at higher levels of compliance [1, 24].

More recent trials are beginning to compare or blend modalities (e.g., additional sprint training versus additional NHE) to target similar modifiable risk factors, reflecting the mixed approaches used by many teams in practice [14, 27]. Beyond NHE and FIFA 11 +, other exercise-based programs have been evaluated and were, therefore, included as separate nodes in the present network. FIFA 11 is the earlier FIFA injury-prevention warm-up program, whereas FIFA 11 + Kids is an age-adapted version developed for younger players. The bounding exercise program (BEP) represents a hamstring-focused approach emphasizing running-related and plyometric loading. However, the evidence base for FIFA 11, FIFA 11 + Kids, and BEP remains smaller than that for NHE and FIFA 11 +, with some nodes supported

by only one or two trials. Because studies evaluate different exercise packages and direct head-to-head comparisons are limited, clinicians still lack clear comparative effectiveness evidence for hamstring injury prevention [6, 14, 27]. Network meta-analysis (NMA) can address this by combining direct and indirect evidence across a network of interventions, enabling estimation of relative effects between any pair of programs and allowing ranking when assumptions are met [28, 29].

While NMA has recently been applied to injury-prevention research in football [30], no study to date has systematically compared a broad range of established and emerging exercise interventions specifically for hamstring injury prevention using exposure-adjusted incidence rate ratios as the primary effect measure. A key assumption underlying network meta-analysis is transitivity, which requires that trials included in different pairwise comparisons are sufficiently comparable with respect to important effect modifiers. In the context of hamstring injury prevention in football, such modifiers may include age, sex, competitive level, baseline injury risk, prior injury history, football exposure, intervention adherence and fidelity, injury definition, and the nature of usual training in the control group. This assumption is particularly important for the present evidence base, because available trials include youth, adult, and veteran players, male and female participants, and teams from different competitive and conditioning environments.

This systematic review and network meta-analysis aimed to compare the relative effectiveness of exercise-based interventions for preventing hamstring injuries in football players using exposure-adjusted incidence rate ratios as the primary effect measure.

Materials and Methods

We conducted a systematic review and network meta-analysis in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [31]. The research protocol was pre-developed before implementation and registered on the PROSPERO International Systematic Review Registration Platform (registration number: CRD420251243576; [https://www.crd.york.ac.uk/PROSPERO/view/CRD420251243576]) to ensure the transparency of the research process and reduce the risk of selective reporting.

Search Strategy

We searched PubMed, Cochrane Library, Embase, and Web of Science (WOS), with the search period extending from the inception of each database until January 5, 2026. No language restrictions were applied during the database search; however, only studies with full texts available in English were eligible for data extraction. The complete search strategy, final search dates, and database-specific queries are provided in Supplementary Table 1. Study identification was limited to these electronic database searches; reference lists and other sources were not additionally searched or consulted to identify eligible studies. We followed the PICOS framework[32] to clarify the research question and compiled all terms utilized in subsequent searches, including both Medical Subject Headings (i.e., MeSH and entry terms).

Selection Criteria

Studies were eligible if they met the following criteria: (1) participants were football players, regardless of competitive level, sex, or age; (2) the intervention was exercise- or training-based and aimed to prevent hamstring injuries; (3) a control group was included; (4) hamstring injury incidence and exposure data were available or could be estimated; and (5) the study was an individual or cluster randomized controlled trial. Detailed eligibility criteria according to the PICOS framework are provided in Supplementary Table 2. Hamstring injury was defined as an injury occurring during training or match play that prevented the player from participating in the next training session or match [33]. Studies were excluded if they involved non-football players, non-exercise interventions, non-randomized designs, unavailable or non-extractable outcome data, outcomes unrelated to hamstring injury, conference abstracts, case reports, retrospective studies, cohort studies, or full texts that could not be obtained. BJH and HL independently screened titles and abstracts and then assessed the full texts of potentially eligible studies. Disagreements were resolved through discussion or adjudication by a third reviewer (XH).

Data Extraction

BJH and HL independently extracted data using a standardized form that was piloted before formal extraction. Disagreements were resolved through discussion or adjudication by a third reviewer (XH). Extracted data included study identification, participant sex and age, intervention and comparator descriptions, program duration and frequency, exposure hours, hamstring injury events, sample size, adherence or implementation-fidelity information, and injury definitions. Intervention and comparator characteristics included the intervention program name/type (e.g., NHE, FIFA 11+, FIFA 11, FIFA 11+Kids, BEP) and the control condition. Injury definitions were extracted and classified as time-loss/Fuller-compatible, broad physical-complaint based, medical-attention based, or unclear/mixed. Definitions were considered time-loss/Fuller-compatible when injuries prevented full participation in subsequent training

or match play, and broad physical-complaint based when posterior-thigh complaints were recorded irrespective of time loss or medical attention. Classification decisions for the restricted injury-definition sensitivity analysis are presented in Supplementary Table 3.

Risk-of-Bias Assessment

Risk of bias was assessed at the study level using the RoB 2 tool [34]. Two reviewers independently judged each trial across five domains: (D1) randomization process, (D2) deviations from intended interventions, (D3) missing outcome data, (D4) measurement of the outcome, and (D5) selection of the reported result. Each domain and the overall judgment were rated as low risk, some concerns, or high risk, with disagreements resolved through discussion and re-checking of the full text and supplementary materials where needed. Judgements were summarized visually using a study-level traffic-light plot and a domain-level summary figure. Study-level risk-of-bias judgements are presented in Supplementary Table 4, with the corresponding traffic-light plot and domain-level summary shown in Supplementary Figs. 1 and 2, respectively. We assessed the certainty of evidence for each primary intervention-versus-usual-training network estimate using a GRADE/CINeMA-informed framework. Six domains were considered: within-study bias, reporting bias, indirectness, imprecision, heterogeneity, and incoherence. Each domain was judged as having no concerns, some concerns, or major concerns, and the overall certainty of evidence was classified as high, moderate, low, or very low. Within-study bias judgments were informed by the RoB 2 assessments of the contributing trials. Imprecision was assessed by considering whether the 95% confidence interval included the null effect and effects of potentially important magnitude in either direction. Heterogeneity was assessed using τ^2, I^2, confidence-interval width, and clinical and methodological diversity. Incoherence was planned to be assessed by comparing direct and indirect evidence; however, it was not estimable, because the network contained no closed loops. Comparison-specific judgments and reasons for reduced certainty are presented in Supplementary Table 5.

Data Synthesis

All extracted data were synthesized using random-effects network meta-analysis (NMA) to compare exercise-based prevention strategies. Each intervention was treated as a separate node: FIFA 11, FIFA 11+, FIFA 11+Kids, Nordic hamstring exercise (NHE), bounding exercise program (BEP), and usual training. The primary effect measure was the incidence rate ratio (IRR), using usual training as the reference node. Their inclusion in the network allows available evidence across intervention types to be compared, but estimates for these sparsely informed nodes require cautious interpretation. Effect estimates were expressed as IRRs with 95% confidence intervals (CIs); IRR values below 1 indicated lower hamstring injury incidence than the comparator. Statistical significance was defined as a two-sided p value < 0.05 or a 95% CI excluding 1.00. All statistical analyses were performed using Stata version 18.0 (StataCorp, College Station, TX, USA). Network meta-analysis was conducted using the Stata network meta-analysis suite for network setup, random-effects estimation, relative-effect plots, treatment ranking, and comparison-adjusted funnel plots. Any mention of the R package netmeta in the protocol documentation was a reporting error; the analyses reported in this manuscript were conducted in Stata.

Usual training was retained as a single reference node to preserve network connectivity, because included trials generally used non-targeted routine training or usual warm-up comparators. Because usual training was not identical across trials, comparator characteristics were examined descriptively and considered when interpreting transitivity and indirect comparisons.

Relative effects versus usual training were presented in forest or relative-effect plots. To describe the intervention hierarchy, we calculated P-scores based on cumulative ranking probabilities [35]. Because lower IRR values indicate lower injury incidence, interventions with lower estimated IRRs were ranked more favorably. P-scores were interpreted alongside effect estimates, 95% CIs, number of trials per node, and certainty of evidence, rather than as stand-alone evidence of superiority. Between-study heterogeneity was assessed using τ^2, I^2, H^2, and Cochran's Q statistic where applicable. I^2 values of approximately 25%, 50%, and 75% were interpreted descriptively as low, moderate, and substantial heterogeneity, respectively. Because τ^2 is scale-dependent, it was interpreted together with I^2, H^2, Q statistics, clinical heterogeneity, and confidence-interval width.

Inconsistency was planned to be assessed using loop-specific inconsistency factors or node-splitting, where closed loops were present. Because the primary IRR network contained no closed loops, formal inconsistency testing was not estimable; indirect estimates were, therefore, interpreted in light of the transitivity assumption. Therefore, indirect comparisons and treatment rankings derived from this network were interpreted as population-level comparative estimates rather than as a definitive universal hierarchy applicable to every football setting or individual player. Small-study effects and publication bias were explored using comparison-adjusted funnel plots, where effect sizes were centered at the comparison-specific pooled effect and plotted

against their standard errors [36]. Egger's regression test was additionally performed for the primary IRR synthesis as an exploratory assessment of funnel-plot asymmetry. Because the number of included studies was small and several nodes were sparsely informed, Egger's test was interpreted as supplementary rather than definitive evidence for or against publication bias.

Given substantial heterogeneity in the primary IRR synthesis, we conducted an exploratory subgroup NMA restricted to male players to explore whether sex-related differences in exposure profiles and baseline risk could contribute to variability in the exposure-adjusted estimates. The male-only subgroup was selected, because 9 of the 11 included trials enrolled male players, preserving a connected network across the main intervention nodes. A separate female-only NMA was not performed, because only two trials enrolled female players, and these trials evaluated different intervention nodes, leaving the female-only evidence base too sparse to support stable network estimates, heterogeneity assessment, or meaningful treatment ranking. The male-only subgroup analysis used the same frequentist random-effects framework and reported IRRs. To address age-related clinical heterogeneity, we also conducted exploratory age-stratified subgroup NMAs. Studies enrolling children, adolescents, youth players, or participants with a mean age below 18 years were classified as the youth/adolescent subgroup. Studies enrolling adult or senior competitive players, or participants with a mean age of 18 years or older, were classified as the adult/senior subgroup. Engebretsen et al.[37] did not report participant age but enrolled players from the top national divisions and was, therefore, classified as adult/senior. Hammes et al.[38] was included in the adult/senior subgroup, because it enrolled male veteran football players; its influence was additionally examined in the sensitivity analysis excluding this trial.

To examine the influence of the veteran-player trial, we conducted a sensitivity analysis excluding Hammes et al.[38], which enrolled male veteran football players with a mean age in the mid-40 s. Resulting IRRs, 95% CIs, residual heterogeneity estimates, and treatment ordering were compared with the primary analysis. To examine injury-definition heterogeneity, we conducted a sensitivity analysis restricted to studies using time-loss or Fuller-compatible hamstring injury definitions. Studies using broad physical-complaint definitions irrespective of time loss or medical attention, or reporting non-time-loss injuries, were excluded. Resulting IRRs, 95% CIs, and P-scores were compared with the primary analysis. Leave-one-out analyses were retained as influence analyses to assess whether any single trial disproportionately affected pooled estimates or the treatment hierarchy.

Results

Study Selection

The database search identified 654 potentially relevant records. After duplicates were removed, titles and abstracts were screened, and 68 records were selected for full-text assessment. After full-text assessment, 11 randomized controlled trials were included in the systematic review and network meta-analysis. The entire literature screening process and the reasons for exclusion at each stage are illustrated in the PRISMA flowchart (Fig. 1). The included trials evaluated exercise-based interventions for hamstring injury prevention in football players and included both individual and cluster randomized designs.

Characteristics of Included Studies

The characteristics of the included studies are summarized in Table 1. The 11 trials were published between 2008 and 2023 and included 9,282 participants. All trials were two-arm controlled studies and enrolled football players across different competitive levels, age groups, and sexes. One included trial [38] enrolled male veteran football players with a mean age in the mid-40 s, which was clinically distinct from the adolescent and younger adult football populations included in the other trials. Intervention nodes included NHE ($n=3$), FIFA 11 + ($n=4$), FIFA 11 ($n=2$), FIFA 11 + Kids ($n=1$), and BEP ($n=1$), with variation in program content and implementation methods. Because usual training served as the common reference node, comparator conditions were extracted in detail. Control groups were variably described as usual warm-up, standard warm-up, regular soccer training, regular training routine, or practice as usual, with differing detail on structured prevention, clinical support, and contamination monitoring. Comparator characteristics are summarized in Supplementary Table 6. Intervention duration, frequency, and content varied substantially. Program duration ranged from approximately 10–13 weeks to 24–39 weeks or approximately 8 months. In most trials, the interventions were delivered one to three times per week, whereas Hammes et al.[38] implemented the FIFA 11 + program during every team training session. This variability in planned and delivered dose was considered an important source of clinical and implementation heterogeneity. Adherence and implementation-fidelity reporting varied across trials. Some studies reported quantitative team- or player-level compliance, whereas others reported completed sessions, program exposure, or qualitative implementation information. Reported adherence ranged from low completion of prescribed exercise volume to high team-level compliance. For example, Al Attar et al.[40]

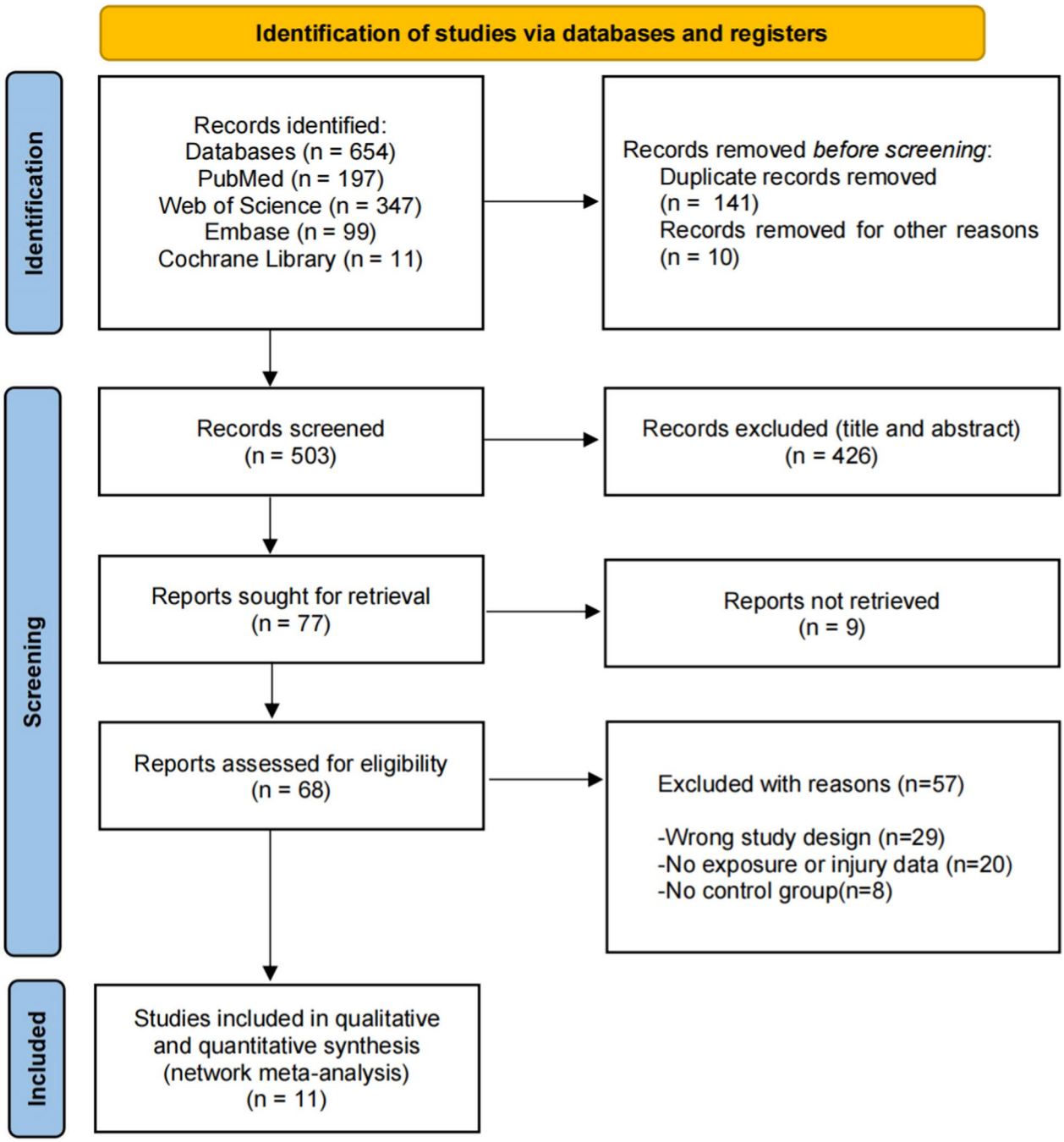


Fig. 1 PRISMA flow diagram of study selection. The diagram shows the numbers of records identified, removed before screening, screened, assessed for eligibility, excluded with reasons, and included in the systematic review and network meta-analysis. PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses

reported high team-level compliance, Engebretsen et al. [37] reported low compliance, and Silvers-Granelli et al. [24] reported a utilization gradient in which higher FIFA 11 + use was associated with lower injury rates. Because adherence metrics were heterogeneous and sparse within intervention nodes, adherence was summarized descriptively rather than modeled quantitatively. Details are presented in Supplementary Table 7. Injury definitions also varied. Seven studies used time-loss or Fuller-compatible definitions and were included in the restricted injury-definition sensitivity analysis [45]. Four studies used broader physical-complaint definitions irrespective

Table 1 Characteristics of the studies included

Reference	Sex	Age (years)	Intervention program	Control program	Intervention protocol	Exposure hours	Injury events	Sample
Hasebe et al. (2019) [39]	M	IG: 16.7 ± 0.5 CG: 16.3 ± 0.6	NHE	Usual training	13 weeks; 2 sessions per week; 25 sessions	IG: 45,374 CG: 28,910	IG: 4 CG: 3	Total:259 IG:156 CG:103
Al Attar et al. (2023) [40]	M	IG:7–13 CG:7–13	FIFA 11 + Kids	Usual training	24 weeks; 2 sessions per week; 48 sessions	IG: 50,120 CG: 42,616	IG: 2 CG: 4	Total:740 IG: 377 CG: 363
Soligard et al. (2008) [23]	F	IG: 15.4 ± 0.7 CG: 15.5 ± 0.7	FIFA 11 +	Usual training	8 months; 2 sessions per week	IG: 49,899 CG: 45,428	IG: 5 CG: 8	Total:1928 IG:1055 CG:837
Steffen et al. (2008) [41]	F	IG: 15.4 ± 0.8 CG: 15.4 ± 0.8	FIFA 11	Usual training	32 weeks; 1–2 sessions per week; 32–64 sessions	IG: 66,423 CG: 65,725	IG: 35 CG: 28	Total:2,020 IG:1,073 CG:947
van de Hoef et al. (2019) [42]	M	IG: 23.8 ± 6.4 CG: 22.2 ± 3.1	BEP	Usual training	39 weeks; 2 sessions per week	IG: 66,423 CG: 65,725	IG: 31 CG: 26	Total:588 IG:305 CG:283
van der Horst et al. (2015)[18]	M	IG: 24.5 ± 3.6 CG: 24.6 ± 4.1	NHE	Usual training	13 weeks; 2 sessions per week; 25 sessions	IG: 30,564 CG: 29,635	IG: 11 CG: 25	Total:648 IG:329 CG:319
Engebretsen et al. (2008) [37]	M	Not reported	NHE	Usual training	10 weeks; 3 sessions per week	IG: 41,856 CG: 40,913	IG: 23 CG: 17	Total:388 IG:193 CG:195
Silvers-Granelli et al. (2015) [24]	M	IG:20.40 ± 1.66 CG: 20.68 ± 1.46	FIFA 11 +	Usual training	4 months; 3 sessions per week	IG: 35,226 CG: 44,212	IG: 16 CG: 55	Total:1525 IG:675 CG:850
Van Beijsterveldt et al. (2012) [43]	M	IG: 24.4 ± 4.1 CG: 25.1 ± 4.3	FIFA 11	Usual Training	33 weeks; 2 sessions per week; 66 sessions	IG: 21,605 CG: 22,647	IG: 207 CG: 220	Total:487 IG:241 CG:246
Owoeye et al. (2014) [44]	M	IG: 17.80 ± 0.94 CG: 17.49 ± 1.1	FIFA 11 +	Usual training	24 weeks; 2 sessions per week; 48 sessions	IG: 51,017 CG: 61,045	IG: 36 CG: 94	Total:416 IG:212 CG:204
Hammes et al. (2014) [38]	M	IG: 45.2 ± 7.7 CG: 43.1 ± 6.5	FIFA 11 +	Usual training	39 weeks; Each training session	IG: 4,172 CG: 2,937	IG: 51 CG: 37	Total:283 IG:158 CG:125

BEP bounding exercise program, *CG* control group, *F* female, *FIFA* Fédération Internationale de Football Association, *IG* intervention group, *M* male, *NHE* Nordic hamstring exercise

of time loss or reported non-time-loss injuries and were excluded from this restricted analysis. Definitions, classifications, and inclusion decisions are presented in Supplementary Table 3. All studies reported hamstring injury events and exposure time, enabling calculation of injury incidence.

The Preventive Effect of Exercise-Based Intervention on Hamstring Injuries

Network Geometry

The primary IRR network included six nodes: NHE, FIFA 11, FIFA 11 +, FIFA 11 + Kids, BEP, and usual training (Fig. 2). Direct evidence was available only for active intervention-versus-usual-training comparisons; no direct active–active comparisons or closed loops were present. The availability of direct comparisons, closed loops, and inconsistency assessment is summarized in Supplementary Table 8. FIFA 11 + versus usual training was the most frequently informed comparison (n = 4). Therefore, active–active estimates were derived indirectly through usual training and were interpreted in light of the transitivity assumption [46]. Edge thickness is proportional to the number of studies informing each direct comparison. The greatest amount of direct evidence was available for comparisons of FIFA 11 + and NHE against usual training, whereas FIFA 11 + Kids and BEP were each supported by a single trial. Despite this imbalance, the network remained connected, allowing all interventions to be analyzed within a single NMA framework.

Ranking of Interventions

FIFA 11 + showed the most consistent preventive signal, with significantly lower hamstring injury incidence than usual training (IRR = 0.55, 95% CI 0.31–0.98). FIFA 11 + Kids showed a favorable but imprecise estimate (IRR = 0.43, 95% CI 0.08–2.32), reflecting limited

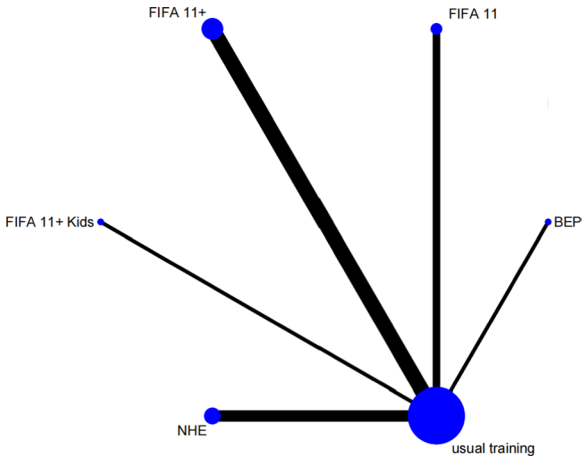


Fig. 2 Network geometry of the primary incidence rate ratio analysis. Each node represents an intervention, with node size proportional to the total sample size contributing to that intervention. Lines represent available direct comparisons, and line thickness is proportional to the number of studies informing each comparison. All active interventions were directly compared with usual training; no direct active–active comparisons or closed loops were present. *BEP* bounding exercise programme, *FIFA* Fédération Internationale de Football Association, *NHE* Nordic hamstring exercise

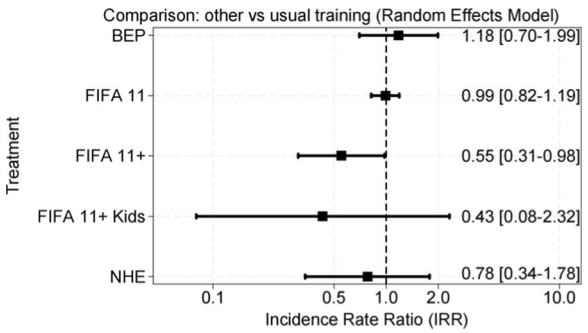


Fig. 3 Relative effects in the primary incidence rate ratio network meta-analysis. Squares represent incidence rate ratio estimates, and horizontal lines represent 95% confidence intervals. The vertical dashed line at 1.00 indicates no difference; values below 1.00 favor the intervention over usual training. *BEP* bounding exercise program, *CI* confidence interval, *FIFA* Fédération Internationale de Football Association, *IRR* incidence rate ratio, *NHE* Nordic hamstring exercise

direct evidence. NHE showed a modest but non-significant favorable estimate (IRR = 0.78, 95% CI 0.34–1.78), whereas FIFA 11 (IRR = 0.99, 95% CI 0.82–1.19) and BEP (IRR = 1.18, 95% CI 0.70–1.99) showed no clear benefit. Relative effects for all pairwise comparisons are presented in Fig. 3. Complete random-effects network meta-analysis estimates are presented in Supplementary Table 9. The league table of all pairwise network comparisons is provided in Supplementary Table 10. Based on P-scores, FIFA 11 + ranked highest (0.84), followed by FIFA 11 + Kids (0.78), NHE (0.56), usual training (0.34), FIFA 11 (0.27), and BEP (0.21). The apparent ranking of FIFA 11 + Kids should be interpreted cautiously, because this node was supported by a single trial in a very young population and had a wide confidence interval. Therefore, its favorable P-score reflects substantial statistical uncertainty rather than reliable evidence of superiority. Cumulative ranking curves and P-score plots are shown in Fig. 4. Between-study heterogeneity in the primary IRR synthesis was substantial ($\tau^2 = 0.16$; $I^2 = 72.44\%$; $H^2 = 3.63$; Cochran's $Q = 31.90$, df = 10, $p < 0.01$). Therefore, although FIFA 11 + showed the most favorable statistically significant estimate versus usual training, the treatment hierarchy should be interpreted cautiously, because heterogeneity may reflect differences in population characteristics, exposure profiles, intervention delivery, adherence, injury definitions, and usual-training comparators.

Exploratory Male-Only Subgroup Analysis

The exploratory male-only subgroup analysis was conducted, because 9 of the 11 included trials enrolled male players, whereas only two trials enrolled female players and evaluated different active intervention nodes. A separate female-only NMA was, therefore, not feasible. The male-only network remained connected but star-shaped, with usual training as the central reference node and no direct active–active comparisons or closed loops (Fig. 5). In the male-only network, FIFA 11 + remained significantly associated with lower hamstring injury incidence versus usual training (IRR = 0.56, 95% CI 0.34–0.91). FIFA 11 + Kids showed a favorable but highly imprecise estimate (IRR = 0.43, 95% CI 0.08–2.32), while NHE showed a non-significant favorable estimate (IRR = 0.78, 95% CI 0.34–1.78). FIFA 11 (IRR = 1.02, 95% CI 0.85–1.21) and BEP (IRR = 1.18, 95% CI 0.70–1.99) showed no clear benefit. Based on P-scores, FIFA 11 + ranked highest (0.82), followed by FIFA 11 + Kids (0.77), NHE (0.54), FIFA 11 (0.34), usual training (0.31), and BEP (0.21). The league table of all pairwise comparisons from the primary analysis is provided in Supplementary Table 10. The full results of the male-only random-effects model are presented in Supplementary Table 11. However, as in the primary analysis, the ranking of FIFA 11 + Kids should be interpreted cautiously, because it was supported by a single trial and had a very wide confidence interval. Relative effects and ranking plots are presented in Fig. 6 and Fig. 7, respectively. Between-study heterogeneity remained substantial in the male-only IRR synthesis ($\tau^2 = 0.18$; $I^2 = 75.35\%$; $H^2 = 4.06$; Cochran's $Q = 29.09$, df = 8, $p < 0.01$), indicating that restriction to male players did not eliminate between-study variability. Therefore,

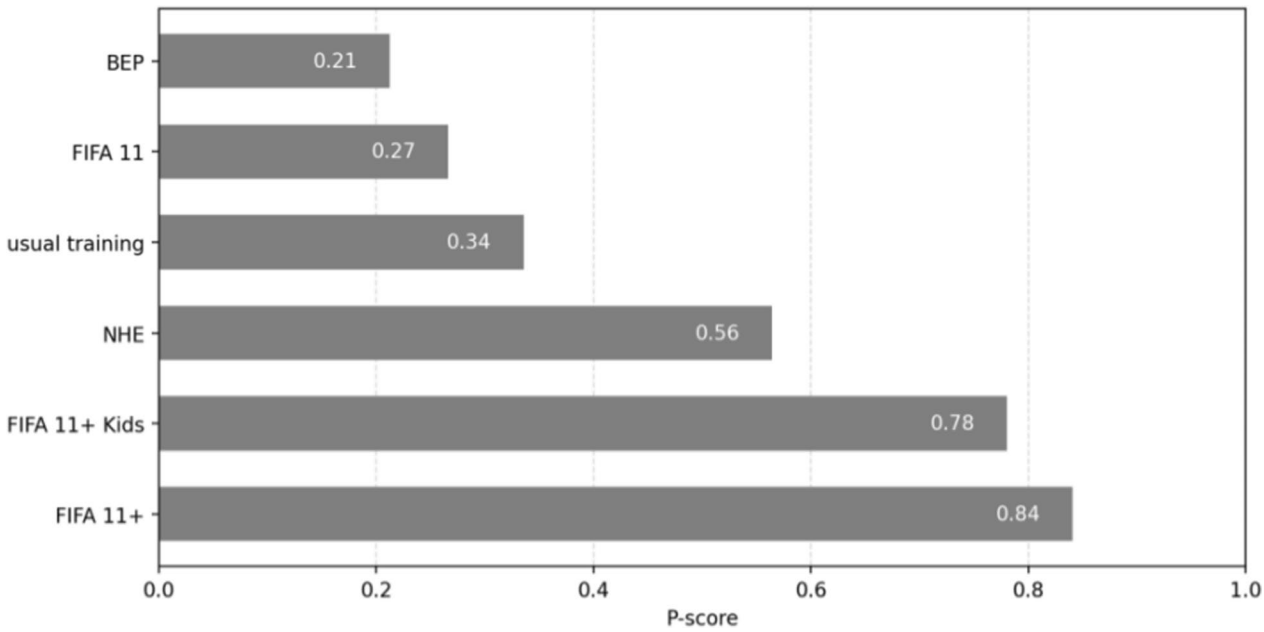


Fig. 4 P-score rankings in the primary network meta-analysis. Bars show the P-score for each intervention, with higher values indicating a more favorable relative ranking. P-scores should be interpreted alongside effect estimates, confidence intervals, evidence certainty, and the amount of supporting evidence. *BEP* bounding exercise program, *FIFA* Fédération Internationale de Football Association, *NHE* Nordic hamstring exercise

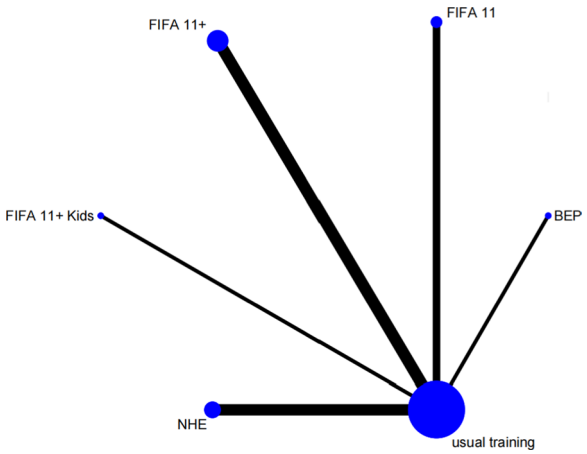


Fig. 5 Network geometry of the male-only subgroup analysis. Node size is proportional to the total sample size for each intervention, and edge thickness is proportional to the number of studies informing each direct comparison. *BEP* bounding exercise program, *FIFA* Fédération Internationale de Football Association, *NHE* Nordic hamstring exercise

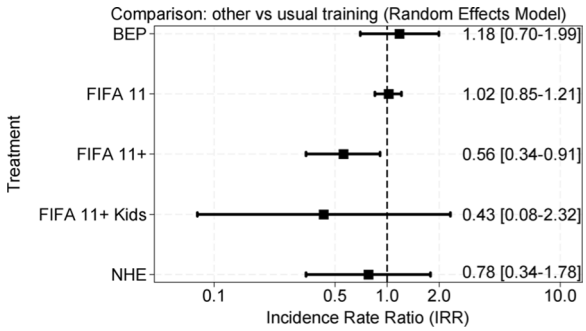


Fig. 6 Relative effects in the male-only subgroup network meta-analysis. Squares represent incidence rate ratio estimates, and horizontal lines represent 95% confidence intervals. The vertical dashed line at 1.00 indicates no difference; values below 1.00 favor the intervention over usual training. *BEP* bounding exercise program, *CI* confidence interval, *FIFA* Fédération Internationale de Football Association, *IRR* incidence rate ratio, *NHE* Nordic hamstring exercise

the male-only results were interpreted as exploratory rather than definitive subgroup-specific rankings.

Age-Stratified Subgroup Analyses

To examine whether age-related clinical heterogeneity influenced the findings, we conducted exploratory age-stratified subgroup NMAs. The youth/adolescent subgroup included five trials enrolling children, adolescents, or youth players. The network remained connected and contained no closed loops; therefore, formal inconsistency testing was not applicable. In this subgroup, FIFA 11 showed a statistically significant reduction in hamstring injury incidence compared with usual training (IRR = 0.38, 95% CI 0.20–0.76, $p = 0.006$). FIFA 11 + (IRR = 0.34, 95% CI 0.06–2.00, $p = 0.230$), FIFA

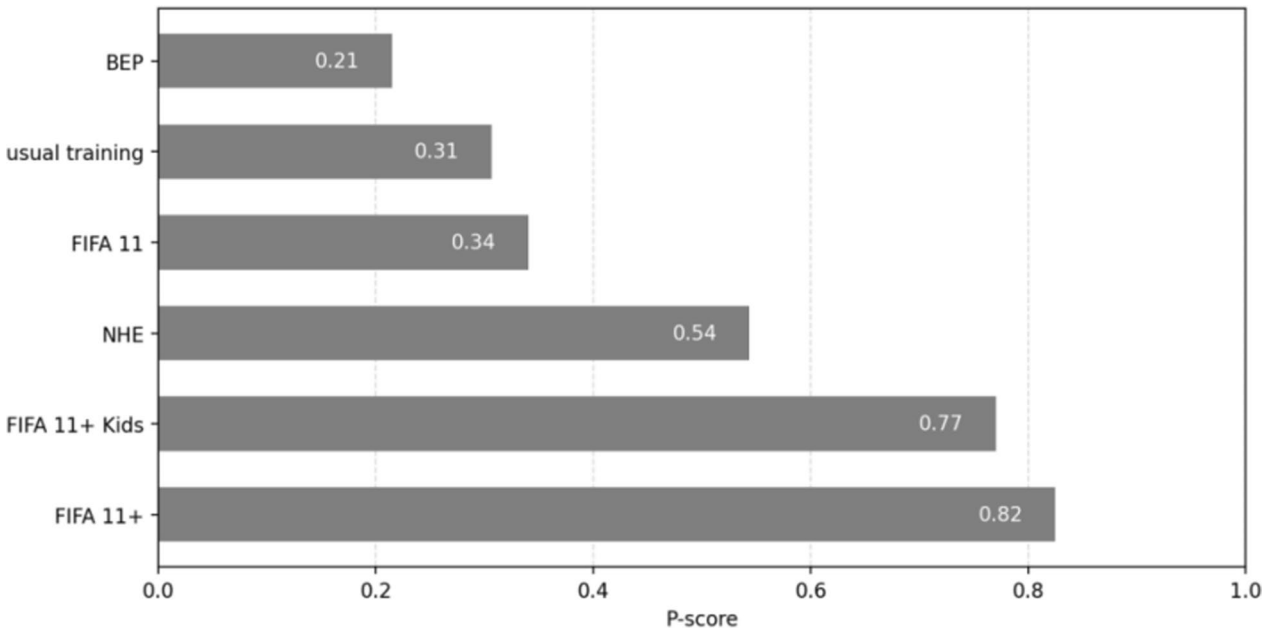


Fig. 7 P-score rankings in the male-only subgroup network meta-analysis. Bars show the P-score for each intervention, with higher values indicating a more favorable relative ranking. These exploratory rankings should be interpreted alongside effect estimates, confidence intervals, evidence certainty, and the amount of supporting evidence. *BEP* bounding exercise program, *FIFA* Fédération Internationale de Football Association, *NHE* Nordic hamstring exercise

11 + Kids (IRR = 0.34, 95% CI 0.06–2.02, $p = 0.240$), and NHE (IRR = 0.69, 95% CI 0.14–3.34, $p = 0.650$) showed favorable but imprecise estimates. Heterogeneity was negligible ($\tau^2 \approx 0.000$; $I^2 \approx 0\%$; $H^2 \approx 1.00$; Cochran's $Q = 0.129$, $p > 0.99$), but ranking remained exploratory, because several nodes were informed by single trials only. The adult/senior subgroup included six trials enrolling adult, senior competitive, or veteran players. This network also remained connected and contained no closed loops. FIFA 11 + had the most favorable point estimate but did not reach statistical significance (IRR = 0.60, 95% CI 0.22–1.64, $p = 0.320$). NHE showed a non-significant favorable estimate (IRR = 0.76, 95% CI 0.25–2.18, $p = 0.610$), whereas FIFA 11 (IRR = 0.99, 95% CI 0.26–3.81, $p = 0.990$) and BEP (IRR = 1.18, 95% CI 0.28–4.93, $p = 0.820$) showed no clear benefit. The best-rank probability was highest for FIFA 11 + (41.8%), followed by NHE (28.4%), FIFA 11 (17.1%), and BEP (11.2%). Heterogeneity was substantial ($\tau^2 = 0.464$; $I^2 = 84.7\%$; $H^2 = 6.52$; Cochran's $Q = 13.03$, $p = 0.001$), indicating important residual variability across adult and veteran football populations. Overall, the age-stratified analyses suggested that the apparent intervention hierarchy was not fully identical across age strata. FIFA 11 showed a statistically significant estimate in the youth/adolescent subgroup, whereas FIFA 11 + had the highest best-rank probability in the adult/senior subgroup but was not statistically significant. Because both age-specific networks were sparse, star-shaped, and dependent on indirect comparisons through usual training, these subgroup results should be interpreted as exploratory rather than definitive age-specific treatment rankings. The full results are presented in Supplementary Table 12.

Risk of Bias, Certainty of Evidence, Publication Bias and Sensitivity Analyses

Visual inspection of the comparison-adjusted funnel plot suggested no clear asymmetry (Fig. 8). Egger's regression test for the primary IRR synthesis also did not indicate statistically significant funnel-plot asymmetry (intercept = −0.9995, 95% CI −3.2568 to 1.2577, $p = 0.343$). However, this assessment was limited by the small number of studies and sparsely informed intervention nodes. Applying the Cochrane RoB 2 tool showed that three studies were judged as high risk of bias, no study was judged as overall low risk, and the remaining eight studies were rated as having some concerns. The study-level traffic-light plot and domain-level summary are presented in Supplementary Figs. 1 and 2, respectively. Comparison-specific certainty ratings for the primary network estimates are presented in Supplementary Table 5. The certainty of evidence was rated as low for FIFA 11 + and FIFA 11 versus usual training and very low for FIFA 11 + Kids, NHE, and BEP versus usual training. The FIFA 11 + estimate was limited mainly by within-study bias, possible reporting bias, indirectness,

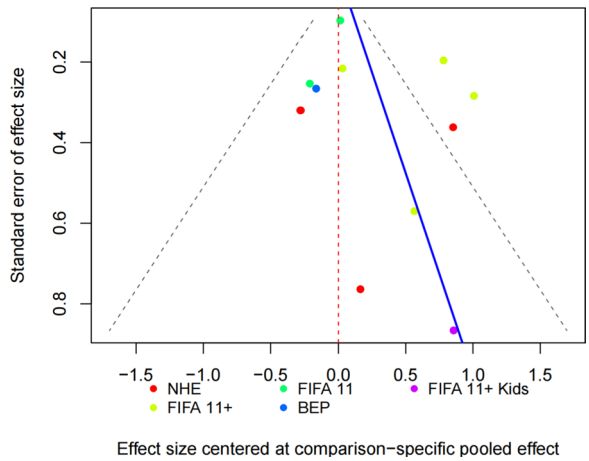


Fig. 8 Comparison-adjusted funnel plot for the primary network meta-analysis. Each point represents an intervention-versus-usual-training comparison and is colored according to the intervention. Effect sizes are centered at the corresponding comparison-specific pooled effect. The red dashed line represents a centered effect of zero, the gray dashed lines represent pseudo-95% confidence limits, and the solid blue line represents the fitted regression line. Interpretation should be considered exploratory because of the small number of studies. *BEP* bounding exercise program, *FIFA* Fédération Internationale de Football Association, *NHE* Nordic hamstring exercise

imprecision, and substantial between-study heterogeneity. The FIFA 11 estimate was limited by sparse direct evidence, within-study bias, possible reporting bias, and clinical and methodological heterogeneity. Certainty was very low for FIFA 11 + Kids, because the estimate was based on a single trial in a very young population and had a very wide confidence interval. Certainty was very low for NHE because of within-study bias, serious imprecision, and substantial heterogeneity, and certainty was very low for BEP, because the estimate was derived from a single study at high risk of bias and had a wide confidence interval. Comparison-specific heterogeneity could not be estimated for single-study nodes, and incoherence could not be assessed, because the network contained no closed loops. Accordingly, the effect estimates, P-scores, and treatment rankings should be interpreted cautiously rather than as robust evidence for definitive practice recommendations. In the injury-definition restricted sensitivity analysis, seven studies using time-loss or Fuller-compatible hamstring injury definitions were included, while four studies using broader physical-complaint definitions or reporting non-time-loss injuries were excluded. The restricted network remained connected but sparse and star-shaped. FIFA 11 + remained directionally favorable but was no longer statistically significant compared with usual training (IRR = 0.65, 95% CI 0.37–1.12). FIFA 11 + Kids showed a favorable but highly imprecise estimate (IRR = 0.43, 95% CI 0.08–2.32), whereas FIFA 11 (IRR = 1.24, 95% CI 0.75–2.03), BEP (IRR = 1.18, 95% CI 0.70–1.99), and NHE (IRR = 1.32, 95% CI 0.71–2.48) showed no clear benefit. Ranking results were unstable, because several nodes were supported by single trials; FIFA 11 + Kids and FIFA 11 + had the highest P-scores (0.83 and 0.82, respectively), but these rankings should be interpreted cautiously given sparse evidence and wide confidence intervals. Between-study heterogeneity remained substantial among repeated FIFA 11 + comparisons ($\tau^2 = 0.147$; $I^2 \approx 70.0\%$). The full results of the injury-definition restricted sensitivity analysis are presented in Supplementary Table 13.

Discussion

Principal Findings

In this IRR-based network meta-analysis of exercise-based hamstring injury-prevention strategies in football, FIFA 11 + showed the most consistent preventive signal compared with usual training (IRR = 0.55, 95% CI 0.31–0.98) and achieved the highest P-score (0.84). FIFA 11 + Kids also received a favorable P-score (0.78), but this estimate was based on a single trial and had a very wide confidence interval. NHE showed a favorable but non-significant estimate (IRR = 0.78, 95% CI 0.34–1.78), whereas FIFA 11 and BEP showed no clear benefit. The hierarchy should be interpreted cautiously, because the network was star-shaped, active–active comparisons were indirect, between-study heterogeneity was substantial, and comparison-specific certainty of evidence ranged from low to very low.

Interpretation and Mechanistic Context

The FIFA 11 + result should be interpreted as an exposure-adjusted comparative signal rather than evidence of mechanistic or causal superiority. Because no direct active–active comparisons were available, the estimated hierarchy does not establish that FIFA 11 + is superior to NHE or other exercise-based approaches [29, 47]. FIFA 11 + has a biologically plausible multi-component rationale, because it incorporates strength, balance, trunk–hip control, landing, cutting, running, and plyometric tasks [48]. However, the included trials did not consistently measure sprint mechanics, biceps femoris long-head fascicle morphology, eccentric knee-flexor strength, pelvic control, or neuromuscular coordination [15]. Mechanistic explanations should, therefore, be regarded as hypothesis-generating. The position of NHE in the population-level hierarchy should likewise not be interpreted as evidence of mechanistic inferiority. NHE is closely aligned with eccentric knee-flexor strength and hamstring architectural adaptations, and its effects may be more pronounced among players with previous

hamstring injury, eccentric strength deficits, short biceps femoris long-head fascicles, or high-risk sprint mechanics [13]. Because the included trials generally evaluated broad team-level populations without stratification by baseline risk, potential benefits in high-risk subgroups may have been diluted. The NHE estimate may also reflect sparse evidence, heterogeneous populations, and variation in adherence and delivered dose [20]. These interpretations are consistent with the anatomical and biomechanical context of hamstring injury. Hamstring strain injuries are strongly associated with high-speed running, particularly during terminal swing, when the hamstrings act eccentrically to decelerate the shank while the hip flexes and the knee extends [10]. The biceps femoris long head is the most frequently injured hamstring muscle, partly because of its biarticular morphology and role in decelerating the lower limb during maximal-speed running [8]. Tissue stress may concentrate near the musculotendinous junction and proximal aponeurosis, where force transmission and local strain gradients are heterogeneous [49]. Preventive exercise may increase tolerance to high-force lengthening contractions or reduce peak strain exposure through improved movement strategies [5]. Lumbopelvic control may also influence hamstring loading, providing a rationale for programs incorporating trunk–pelvis control, balance, deceleration, and landing mechanics [26, 50]. The more consistent FIFA 11 + signal may, therefore, reflect its broad team-level content and standardized delivery rather than superiority in any single mechanism. FIFA 11 + may function as a universal team-wide platform, whereas NHE may be particularly relevant as a targeted component for players with identifiable risk factors [14]. FIFA 11 + is also supported by a larger evidence base than FIFA 11 + Kids or BEP and may be easier to integrate into routine team warm-up practice [2]. Nevertheless, these explanations remain speculative, because the trials did not directly compare active interventions or consistently measure the proposed intermediate mechanisms.

Subgroup Findings

The male-only subgroup broadly mirrored the primary analysis. FIFA 11 + remained significantly associated with lower hamstring injury incidence than usual training (IRR = 0.56, 95% CI 0.34–0.91) and achieved the highest P-score (0.82). FIFA 11 + Kids and NHE remained favorably ranked but imprecise, whereas FIFA 11 and BEP showed no clear benefit. This pattern is biologically plausible, because male football involves substantial high-speed running exposure, and hamstring injuries are commonly associated with sprinting and late-swing loading [10, 12]. However, these mechanisms were not directly assessed in the male-only trials. Substantial heterogeneity remained after restriction to male players ($\tau^2 = 0.18$; $I^2 = 75.35\%$; $H^2 = 4.06$), indicating that sex restriction did not explain most between-study variability. Differences in adherence, coaching quality, program delivery, weekly training load, match congestion, and baseline injury risk may influence the realized effects of prevention programs [51, 52]. The male-only findings should, therefore, be regarded as exploratory and should not be generalized to female players. Age-stratified analyses also suggested that the apparent hierarchy was not identical across age groups. However, both age-specific networks were sparse and star-shaped, and the adult/senior subgroup retained substantial heterogeneity. These analyses should, therefore, be considered exploratory rather than definitive evidence of age-specific comparative effectiveness.

Practical Implications

The practical implications should be interpreted cautiously, because comparison-specific certainty ranged from low to very low and most active–active estimates relied on indirect evidence through usual training. The hierarchy represents a probabilistic population-level summary rather than a definitive prescription for every football setting [47]. Within these limitations, FIFA 11 + may be considered a scalable team-level warm-up option because it showed the most consistent exposure-adjusted signal in the available network [2]. Its value lies in its standardized whole-team format rather than proven mechanistic superiority. A layered approach may be reasonable in practice: FIFA 11 + may provide a universal warm-up platform, while progressive eccentric hamstring strengthening, including NHE-based exercises, may be prioritized for players with previous injury, low eccentric knee-flexor strength, short biceps femoris long-head fascicles, or high-risk sprint profiles [14]. This risk-stratified approach was not directly tested and should be treated as a practical hypothesis. Implementation remains important. program effects are likely to depend on adherence, coaching quality, progression, exercise technique, and delivery of the intended dose within the team schedule [53]. Coaches and clinicians should, therefore, adapt prevention strategies to player characteristics, competitive level, available resources, and usual-training context rather than applying the network ranking as a universal prescription.

Strengths and Limitations

This review integrated multiple exercise-based hamstring injury-prevention strategies within a connected network and used exposure-adjusted IRRs as the primary effect measure. It included 11 randomized controlled trials involving 9282 participants and examined sex, age,

veteran-player influence, comparator heterogeneity, adherence, and injury-definition variability. Several limitations restrict interpretation. First, the network was star-shaped, with all active interventions connected through usual training. No direct active–active comparisons or closed loops were available; therefore, active–active estimates relied on indirect evidence and the transitivity assumption, and formal incoherence testing was not possible. Second, several nodes were sparsely informed. FIFA 11 + Kids was supported by a single trial in a very young population and produced a highly imprecise estimate (IRR = 0.43, 95% CI 0.08–2.32) [40]. Its favorable P-score should not be interpreted as reliable evidence of superiority. BEP was also supported by only one trial. Third, the trials did not consistently measure mechanistic outcomes or stratify results by previous injury, eccentric strength, fascicle length, sprint mechanics, or other risk markers [14]. Population-level estimates may, therefore, obscure subgroup-specific effects, including possible benefits of targeted NHE among high-risk players [13]. Fourth, clinical and methodological heterogeneity was substantial. Participants ranged from children to veteran players, while interventions differed in duration, frequency, adherence, injury definitions, and comparator conditions. The veteran-player trial was particularly distinct from the remaining evidence base. In the sensitivity analysis excluding Hammes et al.[38], the FIFA 11 + estimate became stronger (IRR = 0.434, 95% CI 0.296–0.636) and residual heterogeneity decreased ($\tau^2 = 0.0305$; $I^2 = 26.6\%$). Full results are presented in Supplementary Table 14. Nevertheless, the subgroup and sensitivity analyses were exploratory. Fifth, sex-specific evidence was limited. Only two trials enrolled female players, and they evaluated different intervention nodes. A connected female-only network could, therefore, not be generated, and the male-only findings should not be generalized to female football populations. Sixth, usual-training comparators and injury definitions varied. Usual training included routine warm-up, standard training, or practice as usual, with differences in clinical support and contamination monitoring. In the sensitivity analysis restricted to time-loss or Fuller-compatible injury definitions, FIFA 11 + remained directionally favorable but was no longer statistically significant, suggesting that outcome definitions may affect event counts and comparative estimates [45]. Finally, risk of bias and reporting bias reduced confidence. No trial was judged to be at overall low risk of bias; eight trials had some concerns and three were rated as high risk (Supplementary Table 4 and Supplementary Figs. 1 and 2). Common concerns involved randomization, missing outcome data, and deviations from intended interventions. Under the GRADE/CINeMA-informed assessment, certainty was low for FIFA 11 + and FIFA 11 versus usual training and very low for FIFA 11 + Kids, NHE, and BEP versus usual training (Supplementary Table 5). Although the funnel plot and Egger's regression did not indicate clear asymmetry, the analysis included only 11 trials and several sparsely informed nodes; absence of statistically significant asymmetry should not be interpreted as evidence that reporting bias was absent.

Future Directions

Future trials should provide transparent reporting of adherence, progression, intervention dose, and program fidelity. Direct head-to-head comparisons—particularly FIFA 11 + versus NHE and FIFA 11 + with versus without additional NHE—would reduce reliance on indirect evidence and improve the stability of comparative estimates [36]. Trials should also integrate injury surveillance with mechanism-informed outcomes, including eccentric knee-flexor strength, biceps femoris long-head fascicle morphology, sprint mechanics, and fatigue resistance. Risk-stratified designs could determine whether players with previous injury or identifiable strength, architectural, or sprint-related risk factors benefit from targeted eccentric training beyond a universal warm-up program [12, 17]. Adequately powered studies are needed in female, youth, adult, and veteran populations, with consistent exposure measurement, injury definitions, adherence reporting, and intervention-dose documentation. Future studies should also account for contextual effect modifiers such as match congestion, competitive standard, seasonal timing, and baseline sprint exposure through stratification or prespecified analyses [54].

Conclusion

In this IRR-based network meta-analysis, FIFA 11 + showed the most consistent exposure-adjusted association with reduced hamstring injury incidence compared with usual training. However, this finding should be interpreted cautiously because the evidence base included clinically diverse football populations, varied program duration and delivery frequency, indirect active–active comparisons through usual training, substantial between-study heterogeneity, and moderate-to-high risk of bias across the included studies. Therefore, the observed treatment hierarchy should be regarded as a low-to-very-low-certainty, population-level summary of the currently available evidence rather than as a definitive ranking or a basis for firm recommendations. More definitive clinical recommendations will require well-reported, adequately powered head-to-head trials with clearer intervention-dose reporting, consistent exposure and injury definitions, transparent adherence and implementation-fidelity reporting, and age- or population-specific analyses.

Abbreviations

HSI	Hamstring strain injury
HI	Hamstring injury
IG	Intervention group
CG	Control group
M	Male
F	Female
FIFA	Fédération Internationale de Football Association
IRR	Incidence rate ratio
NHE	Nordic hamstring exercise
BEP	Bounding exercise programme

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s40798-026-01087-w.

Supplementary Material 1.

Supplementary Material 2.

Supplementary Material 3.

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Not applicable.

Author Contributions
BJH and XH designed the study. BJH, XH, HL, and ZYY collected the data. BJH, XH, and HL analyzed and interpreted the data. BJH and JF drafted, reviewed, and edited the manuscript. All authors read and approved the final version of the manuscript.

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Data Availability
All data analyzed in this study were obtained from previously published studies cited in the reference list. The extracted dataset used for the network meta-analysis and the Stata analysis code used to reproduce the primary analysis, subgroup analyses, sensitivity analyses, ranking outputs, Egger's test, and figure generation are provided in the supplementary materials. Additional materials are available from the corresponding author upon reasonable request.

Declarations

Ethics Approval and Consent to Participate
Not applicable.

Consent for Publication
Not applicable.

Competing Interest
BinJie Hong, Xin Hong, Hao Lin, ZiYi Yang, Jun Feng declare that they have no financial or non-financial competing interests that could be perceived to influence the work reported in this paper.

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